

Ritesh Pratap Singh | MCA (GenAI)

Assignment 1 : Java

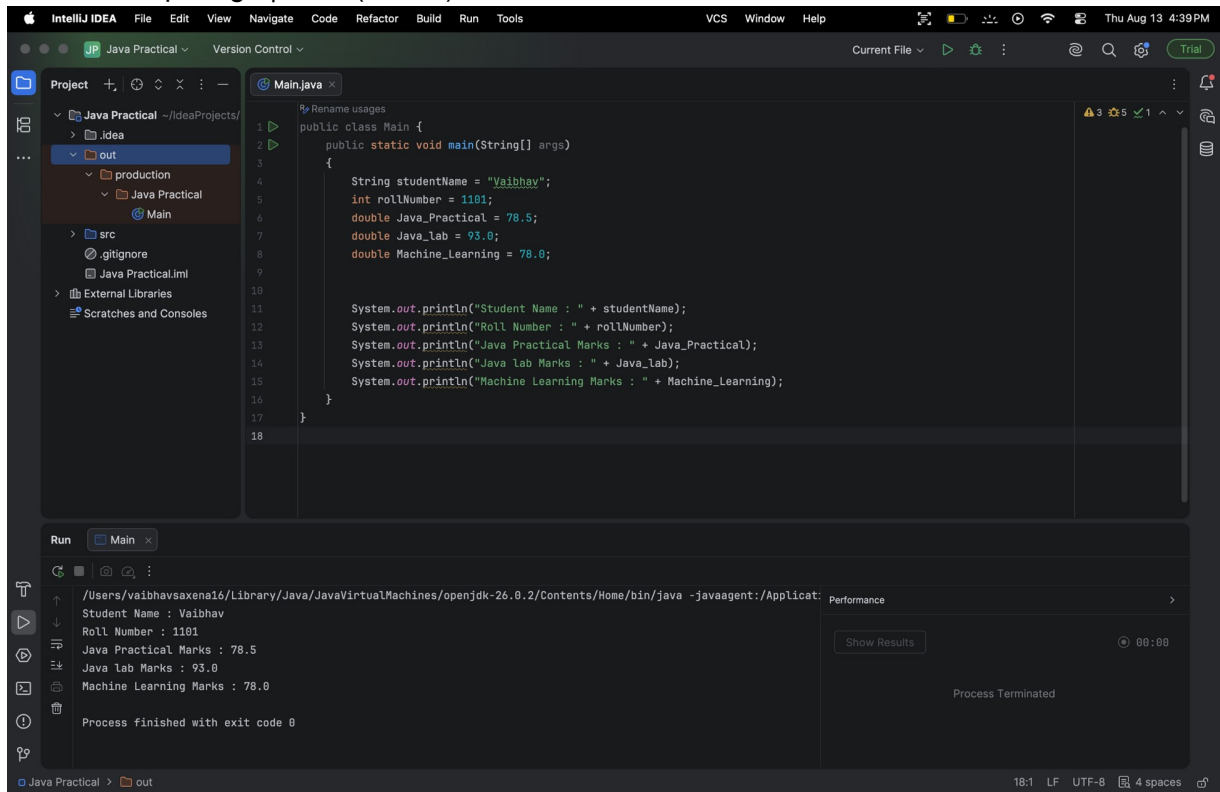
Task 1 :

Write a program that declares variables for a student's name, roll number, and marks in three subjects. Assign values and print them. Ensure all variable names follow Java identifier rules.

Solution -

```
public class StudentInfo {  
    public static void main(String[] args)  
    {  
        String studentName = "Vaibhav";  
        int rollNumber = 1101;  
        double Java_Practical = 78.5;  
        double Java_lab = 93.0;  
        double Machine_Learning = 78.0;  
  
        System.out.println("Student Name : " + studentName);  
        System.out.println("Roll Number : " + rollNumber);  
        System.out.println("Java Practical Marks : " + Java_Practical);  
        System.out.println("Java lab Marks : " + Java_lab);  
        System.out.println("Machine Learning Marks : " + Machine_Learning);  
    }  
}
```

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Task 2 :

Write a Java program that:

- Stores an integer value for temperature in Celsius.
- Converts it to Fahrenheit using the formula $F = (C \times 9/5) + 32$.
- Displays both Celsius and Fahrenheit values.

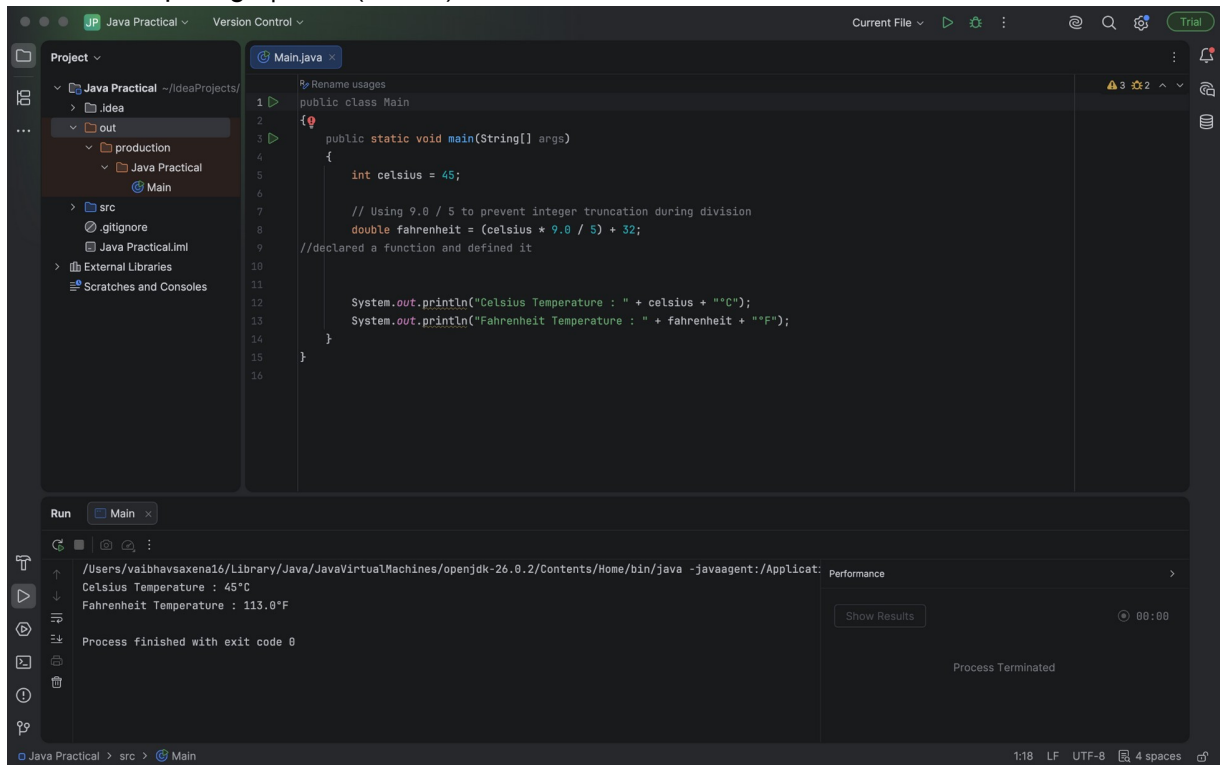
Solution -

```
public class TemperatureConverter
{
    public static void main(String[] args)
    {
        int celsius = 45;

        // Using 9.0 / 5 to prevent integer truncation during division
        double fahrenheit = (celsius * 9.0 / 5) + 32;
        //declared a function and defined it

        System.out.println("Celsius Temperature : " + celsius + "°C");
        System.out.println("Fahrenheit Temperature : " + fahrenheit + "°F");
    }
}
```

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Task 3 :

Accept two integers from the user and perform addition, subtraction, multiplication, division, and modulus. Display results for each operation.

Solution -

```
import java.util.Scanner;
public class ArithmeticOperations {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int num1 = scanner.nextInt();

        System.out.print("Enter second integer: ");
        int num2 = scanner.nextInt();

        System.out.println("Addition (+): " + (num1 + num2));
        System.out.println("Subtraction (-): " + (num1 - num2));
        System.out.println("Multiplication (*): " + (num1 * num2));

        if (num2 != 0) {
            System.out.println("Division (/): " + ((double) num1 / num2));
            System.out.println("Modulus (%): " + (num1 % num2));
        } else {
            System.out.println("Division & Modulus: Cannot divide by zero.");
        }

        scanner.close();
    }
}
```

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The screenshot displays an IDE window for a Java project named "Java Practical". The main editor shows the file "Main.java" with the following code:

```
2 public class Main
4     public static void main(String[] args)
6     {
9         System.out.print("Enter first integer: ");
10        int num1 = scanner.nextInt();
11
12
13        System.out.print("Enter second integer: ");
14        int num2 = scanner.nextInt();
15
16
17        System.out.println("Addition (+): " + (num1 + num2));
18        System.out.println("Subtraction (-): " + (num1 - num2));
19        System.out.println("Multiplication (*): " + (num1 * num2));
20
21
22        if (num2 != 0)
23        {
24            System.out.println("Division (/): " + ((double) num1 / num2));
25            System.out.println("Modulus (%): " + (num1 % num2));
26        }
27        else
28        {
29            System.out.println("Division & Modulus: Cannot divide by zero.");
30        }
31    }
32 }
```

The Run window shows the execution output:

```
Enter first integer: 7
Enter second integer: 14
Addition (+): 21
Subtraction (-): -7
Multiplication (*): 98
Division (/): 0.5
Modulus (%): 7
Process finished with exit code 0
```

The Run window also displays performance metrics for the application:

- Performance
- CPU
- Heap Memory

The status bar at the bottom indicates the file encoding is UTF-8 and the line length is 33:9.

Task 4 :

Write a program to check if a person is eligible to vote.

- Input: age and citizenship status (true or false).
- Use logical operators (&&, ||) to decide eligibility.

Solution :

```
import java.util.Scanner;

public class VotingEligibility
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = scanner.nextInt();

        System.out.print("Are you a citizen? (true/false): ");
        boolean isCitizen = scanner.nextBoolean();

        // Both conditions must be true to vote
        if (age >= 18 && isCitizen)
        {
            System.out.println("Result: Eligible to vote.");
        } else
        {
            System.out.println("Result: NOT eligible to vote.");
        }

        scanner.close();
    }
}
```

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The screenshot displays an IDE interface with a project named "Java Practical". The project structure includes a "src" directory containing a "Main.java" file. The code in "Main.java" is as follows:

```
2  
3  
4 public class Main  
5 {  
6     public static void main(String[] args)  
7     {  
8         Scanner scanner = new Scanner(System.in);  
9  
10  
11         System.out.print("Enter age: ");  
12         int age = scanner.nextInt();  
13  
14  
15         System.out.print("Are you a citizen? (true/false): ");  
16         boolean isCitizen = scanner.nextBoolean();  
17  
18  
19         // Both conditions must be true to vote  
20         if (age >= 18 && isCitizen)  
21         {  
22             System.out.println("Result: Eligible to vote.");  
23         } else
```

The "Run" tab at the bottom shows the execution output:

```
/Users/vaibhavsaxena16/Library/Java/JavaVirtualMachines/openjdk-26.0.2/Contents/Home/bin/java -javaagent:/Applicat:  
Enter age: 21  
Are you a citizen? (true/false): true  
Result: Eligible to vote.  
Process finished with exit code 0
```

The "Performance" tab on the right shows the execution time as 00:00. The status bar at the bottom indicates the file is "Main.java" in the "src" directory, using "UTF-8" encoding with "4 spaces" for indentation.

Task 5 :

Write a program that:• Takes two numbers and an operator symbol (+, -, *, /) from the user. Uses if-else statements to perform the appropriate operation.

Solution :

```
import java.util.Scanner;

public class SimpleCalculator {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = scanner.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = scanner.nextDouble();

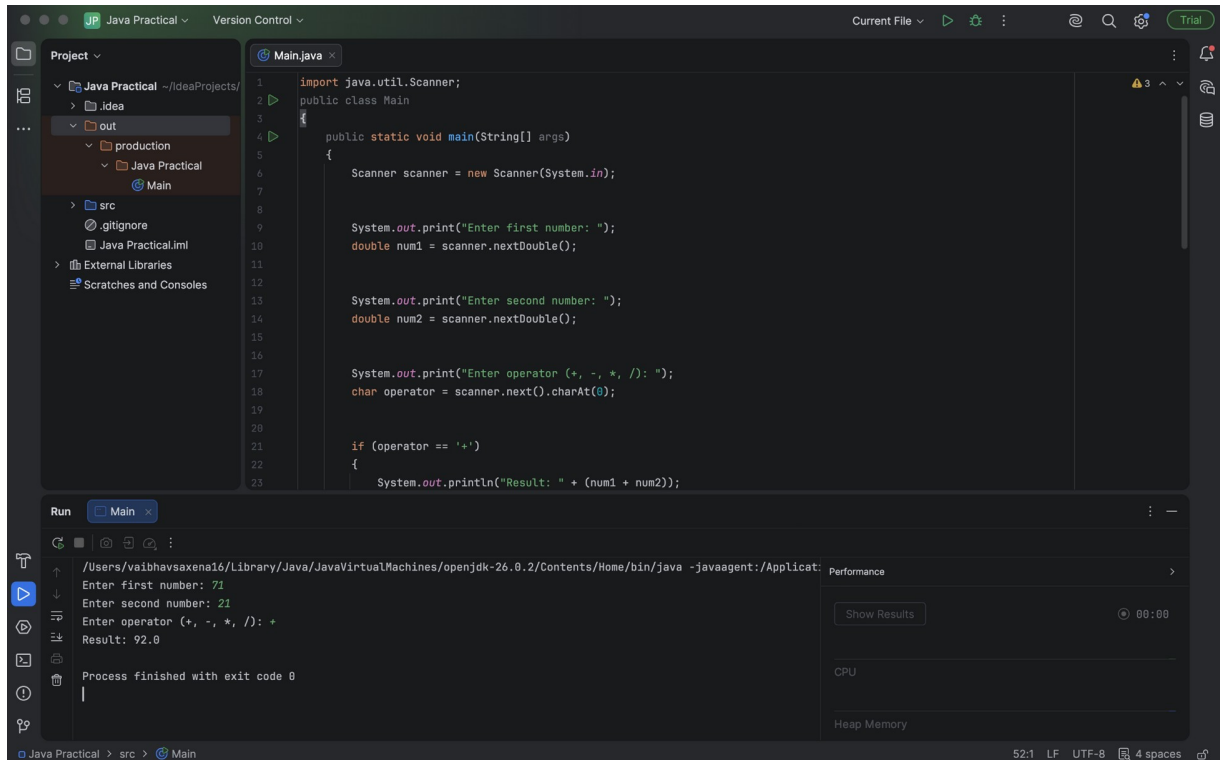
        System.out.print("Enter operator (+, -, *, /): ");
        char operator = scanner.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result: " + (num1 + num2));
        } else if (operator == '-') {
            System.out.println("Result: " + (num1 - num2));
        } else if (operator == '*') {
            System.out.println("Result: " + (num1 * num2));
        } else if (operator == '/') {
            if (num2 != 0) {
                System.out.println("Result: " + (num1 / num2));
            } else {
                System.out.println("Error: Division by zero is not allowed.");
            }
        } else {
            System.out.println("Error: Invalid operator entered.");
        }
    }
}
```

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```
}
```

```
scanner.close(); }}
```



Task 6 :

Ask the user for a number and print its multiplication table from 1 to 10 using a for loop.

Solution :

```
import java.util.Scanner;
```

```
public class MultiplicationTable {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
```

```
        int number = scanner.nextInt();
```

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```
System.out.println("--- Multiplication Table for " + number + " ---");  
for (int i = 1; i <= 10; i++) {  
    System.out.println(number + " x " + i + " = " + (number * i));  
}  
  
scanner.close();  
}  
}
```

The screenshot shows an IDE window titled 'Java Practical'. The left sidebar displays the project structure with folders like 'idea', 'out', 'production', and 'src'. The main editor area shows the 'Main.java' file with the following code:

```
1 import java.util.Scanner;  
2  
3  
4 public class Main  
5 {  
6     public static void main(String[] args)  
7     {  
8         Scanner scanner = new Scanner(System.in);  
9  
10  
11         System.out.print("Enter a number: ");  
12         int number = scanner.nextInt();  
13  
14  
15         System.out.println("--- Multiplication Table for " + number + " ---");  
16         for (int i = 1; i <= 10; i++) {  
17             System.out.println(number + " x " + i + " = " + (number * i));  
18         }  
19  
20  
21         scanner.close();  
22     }  
23 }
```

The bottom panel shows the 'Run' tab with the output of the program. The output is as follows:

```
--- Multiplication Table for 71 ---  
71 x 1 = 71  
71 x 2 = 142  
71 x 3 = 213  
71 x 4 = 284  
71 x 5 = 355  
71 x 6 = 426  
71 x 7 = 497  
71 x 8 = 568  
71 x 9 = 639
```

The right sidebar shows the 'Performance' tab with a 'Show Results' button and a timer showing '00:00'. The status bar at the bottom indicates the file is 'Main.java' in the 'src' directory, with a 'main' method selected. The status bar also shows '18:5', 'LF', 'UTF-8', and '4 spaces'.

Task 7 :

Write a program that uses a while loop to find the sum of all even numbers between 1 and 50.

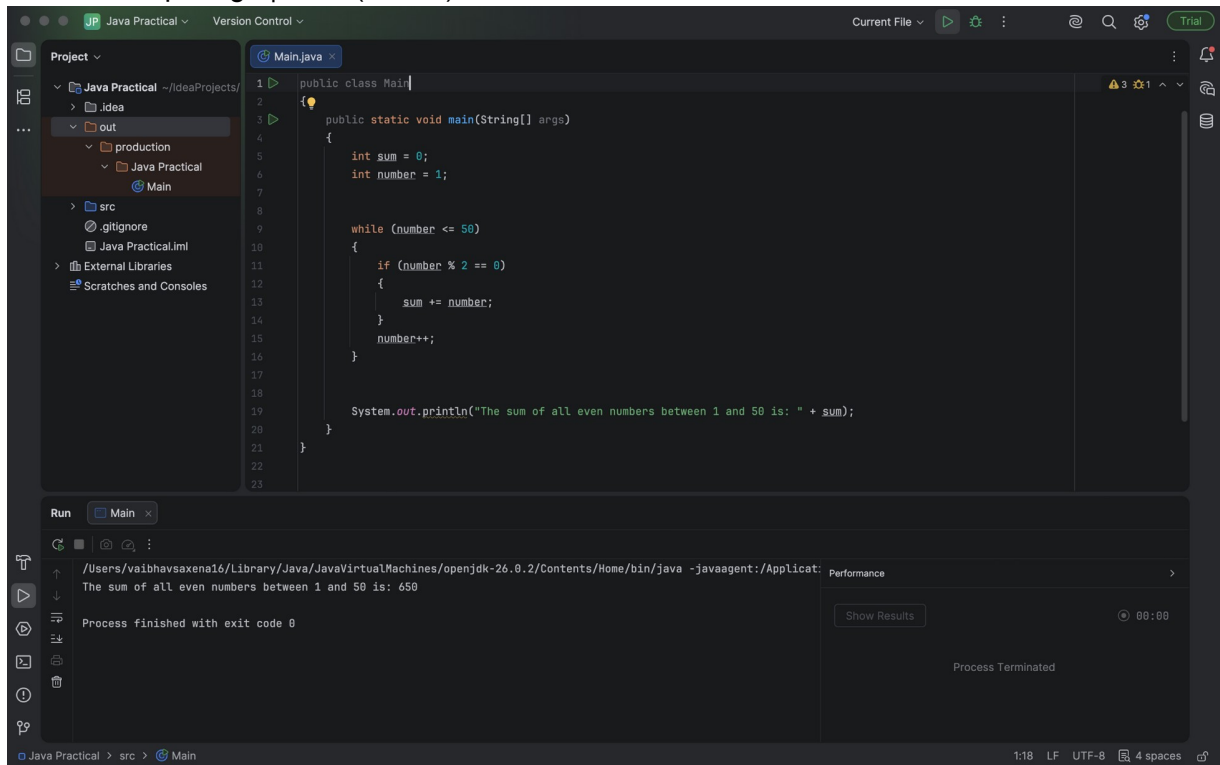
Solution :

```
public class SumOfEvenNumbers
{
    public static void main(String[] args)
    {
        int sum = 0;
        int number = 1;

        while (number <= 50)
        {
            if (number % 2 == 0)
            {
                sum += number;
            }
            number++;
        }

        System.out.println("The sum of all even numbers between 1 and 50 is: " + sum);
    }
}
```

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Task 8 :

Write a program that:

- Reads marks of 5 students into an integer array.
- Prints each student's marks.

Solution :

```
import java.util.Scanner;

public class StudentMarksArray
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);
        int[] marks = new int[5];

        System.out.println("Enter marks for 5 students:");
        for (int i = 0; i < marks.length; i++)
        {
            System.out.print("Student " + (i + 1) + ": ");
            marks[i] = scanner.nextInt();
        }

        System.out.println("\n--- Displaying Stored Marks ---");
        for (int i = 0; i < marks.length; i++)
        {
            System.out.println("Student " + (i + 1) + " Marks: " + marks[i]);
        }

        scanner.close();
    }
}
```

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The screenshot displays an IDE interface with a project named 'Java Practical'. The project structure on the left includes 'out', 'production', and 'src' folders, with 'Main' as the selected class. The main editor shows the code for 'Main.java', which is a Java program that prompts the user to enter marks for 5 students, stores them in an array, and then displays them. The code is as follows:

```
2  
3  
4 public class Main  
5 {  
6     public static void main(String[] args)  
7     {  
8         Scanner scanner = new Scanner(System.in);  
9         int[] marks = new int[5];  
10  
11  
12         System.out.println("Enter marks for 5 students:");  
13         for (int i = 0; i < marks.length; i++)  
14         {  
15             System.out.print("Student " + (i + 1) + ": ");  
16             marks[i] = scanner.nextInt();  
17         }  
18  
19  
20         System.out.println("\n--- Displaying Stored Marks ---");  
21         for (int i = 0; i < marks.length; i++)  
22         {  
23             System.out.println("Student " + (i + 1) + " Marks: " + marks[i]);  
24         }  
25     }  
26 }
```

The 'Run' tab at the bottom shows the execution output, which matches the code's logic:

```
--- Displaying Stored Marks ---  
Student 1 Marks: 52  
Student 2 Marks: 66  
Student 3 Marks: 43  
Student 4 Marks: 88  
Student 5 Marks: 99  
  
Process finished with exit code 0
```

On the right side of the 'Run' tab, there is a 'Performance' section with a 'Show Results' button and a timer showing '00:00'. Below this, there are sections for 'CPU' and 'Heap Memory' usage, each with a progress bar.

The status bar at the bottom indicates the current file is 'Main.java' in the 'src' folder, with a line number of 4:18, LF line endings, UTF-8 encoding, and 4 spaces for indentation.

Task 9 :

Write a program to read 10 integers into an array and find the maximum value using a loop.

Solution :

```
import java.util.Scanner;

public class FindMaximumInArray
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);
        int[] numbers = new int[10];

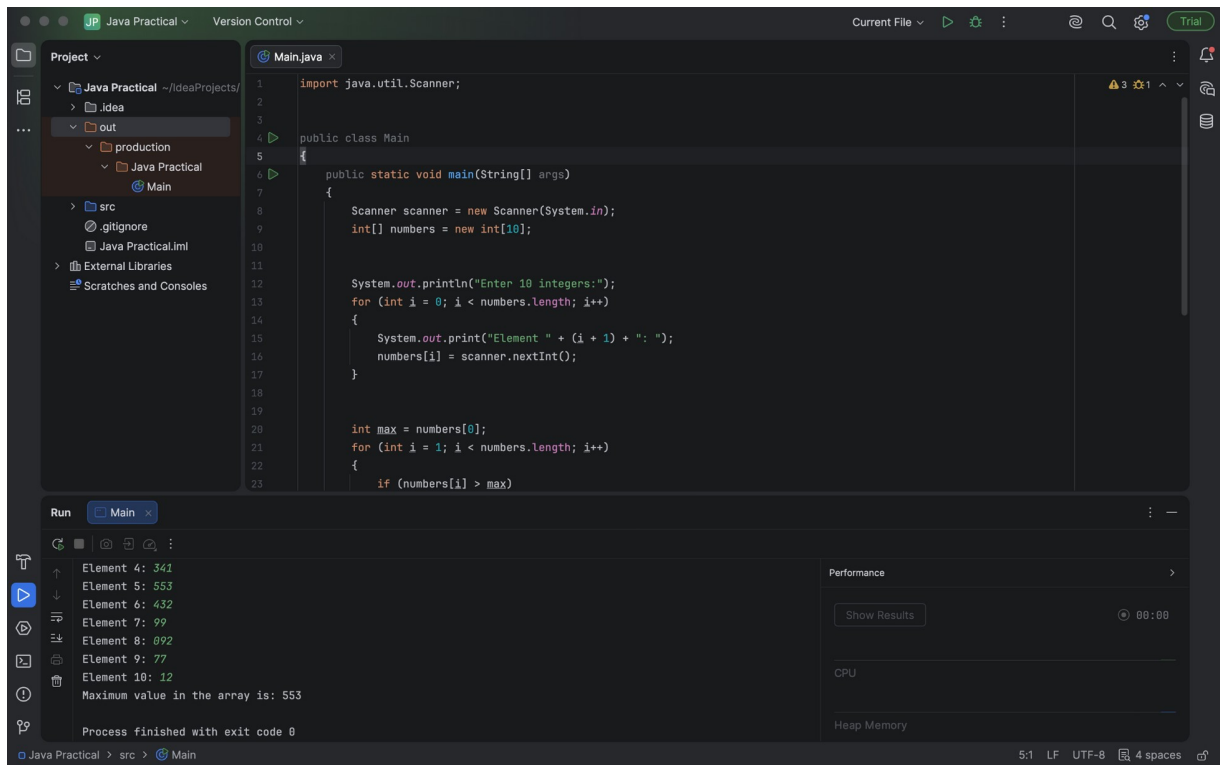
        System.out.println("Enter 10 integers:");
        for (int i = 0; i < numbers.length; i++)
        {
            System.out.print("Element " + (i + 1) + ": ");
            numbers[i] = scanner.nextInt();
        }

        int max = numbers[0];
        for (int i = 1; i < numbers.length; i++)
        {
            if (numbers[i] > max)
            {
                max = numbers[i];
            }
        }

        System.out.println("Maximum value in the array is: " + max);
    }
}
```

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```
        scanner.close();
    }
}
```



The screenshot displays an IDE interface with a project named 'Java Practical'. The 'Project' sidebar on the left shows the file structure, including 'out', 'production', 'Java Practical', and 'Main'. The 'Main.java' file is open in the editor, showing the following code:

```
1 import java.util.Scanner;
2
3
4 public class Main
5 {
6     public static void main(String[] args)
7     {
8         Scanner scanner = new Scanner(System.in);
9         int[] numbers = new int[10];
10
11
12         System.out.println("Enter 10 integers:");
13         for (int i = 0; i < numbers.length; i++)
14         {
15             System.out.print("Element " + (i + 1) + ": ");
16             numbers[i] = scanner.nextInt();
17         }
18
19
20         int max = numbers[0];
21         for (int i = 1; i < numbers.length; i++)
22         {
23             if (numbers[i] > max)
```

The 'Run' console at the bottom shows the output of the program:

```
Element 4: 341
Element 5: 553
Element 6: 432
Element 7: 99
Element 8: 892
Element 9: 77
Element 10: 12
Maximum value in the array is: 553
Process finished with exit code 0
```

The 'Performance' panel on the right shows a 'Show Results' button and a timer at 00:00. The status bar at the bottom indicates the file is 'Main.java' in the 'src' directory, using 'UTF-8' encoding with '4 spaces' for indentation.

Task 10 :

Write a program that:

- Reads marks of 5 subjects into an array.
- Calculates the average marks.
- Uses if-else to assign grades:

o ≥ 90 : A

o ≥ 75 : B o ≥ 50 : C

o Else: Fail

Solution :

```
import java.util.Scanner;

public class StudentGradeSystem
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);
        double[] marks = new double[5];
        double totalSum = 0;

        System.out.println("Enter marks for 5 subjects:");
        for (int i = 0; i < marks.length; i++)
        {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = scanner.nextDouble();
            totalSum += marks[i];
        }

        double average = totalSum / 5.0;
        String grade;
```

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```
    if (average >= 90)
    {
        grade = "A";
    } else if (average >= 75)
    {
        grade = "B";
    } else if (average >= 50)
    {
        grade = "C";
    } else {
        grade = "Fail";
    }

    System.out.println("\nAverage Marks: " + average);
    System.out.println("Grade: " + grade);

    scanner.close();
}
}
```

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The screenshot displays an IDE with a Java project named "Java Practical". The main editor shows the file `Main.java` with the following code:

```
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner scanner = new Scanner(System.in);
7         double[] marks = new double[5];
8         double totalSum = 0;
9
10        System.out.println("Enter marks for 5 subjects:");
11        for (int i = 0; i < marks.length; i++)
12        {
13            System.out.print("Subject " + (i + 1) + ": ");
14            marks[i] = scanner.nextDouble();
15            totalSum += marks[i];
16        }
17
18        double average = totalSum / 5.0;
19        String grade;
20
21        if (average >= 90)
22        {
23
```

The Run console shows the execution output:

```
Enter marks for 5 subjects:
Subject 1: 112
Subject 2: 12
Subject 3: 223
Subject 4: 43
Subject 5: 22
Average Marks: 82.4
Grade: B
1
```

The Performance tab is also visible, showing a "Show Results" button and a timer at 00:00. The status bar at the bottom indicates the file is at line 3, column 18, using UTF-8 encoding with 4 spaces.